

E-Commerce Customer Analytics & RFM Segmentation

Customer Behavior, Revenue Performance & Business Insights

RFM CLUSTERS [R1-F5-M5] 98.4%

- Champions — [High Value]
- Loyal Customers — [High Freq]
- At-Risk — [Low Recency]

▲ Segment Delta: +14.2% MoM

PREPARED BY

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TOOLS USED: Excel • Power Query • MySQL • Python • Statistics • Power BI

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Executive Summary

A decision-focused summary of the analysis

REVENUE	ORDERS	CUSTOMERS	PRODUCTS	AOV
₹17.37M	36,969	5,878	4,631	₹469.98

BUSINESS OBJECTIVE

Evaluate transaction data to understand revenue performance, customer behavior, customer value, product performance and geographic concentration—and convert those findings into practical retention, recovery and growth priorities.

KEY FINDINGS

Finding	Evidence	Business meaning
Customer concentration	633 high-value customers generate ~65.3% of customer revenue.	Retention of high-value relationships is financially critical.
RFM concentration	Champions contribute ~68% of customer revenue.	Champions should receive the strongest retention investment.
Market concentration	UK contributes ~83% of total revenue (~₹14.4M).	The strongest market is also the largest geographic risk.
Uneven customer value	Mean revenue ~₹2,955 vs median ~₹868.	The typical customer is very different from the average customer.

PRIORITY ACTIONS

Priority	Action	Expected business effect
01	Protect Champions and high-value customers	Reduce disproportionate revenue-at-risk.
02	Win back At Risk customers	Recover historical customer value.
03	Develop secondary markets	Improve revenue resilience.
04	Monitor product concentration	Reduce assortment / availability risk.

Executive conclusion

The clearest opportunity is to improve the durability and value of existing customer relationships while reducing dependence on a small customer cohort and one dominant market.

Data & Analytical Approach

A complete, traceable pipeline from raw transactions to insights

SOURCE DATA

Item	Actual project detail
File	online_retail_II.xlsx
Worksheets	Year 2009-2010: 525,461 rows Year 2010-2011: 541,910 rows
Overlap	01-Dec-2010 to 09-Dec-2010; overlap in the second worksheet removed before combining
Final dataset	779,425 analysis-ready transaction records

END-TO-END WORKFLOW

EXCEL Source	POWER QUERY Clean	MYSQL / SQL Validate	PYTHON Analyze	POWER BI Communicate	INSIGHTS Decide
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WHAT WAS DONE AT EACH STAGE

Stage	Work performed
Power Query	Removed exact duplicates, missing Customer IDs, Quantity ≤ 0 and Price ≤ 0 ; checked Description/Country; converted InvoiceDate; created Revenue = Quantity $\times$ Price; prepared SQL-ready datetime.
MySQL + SQL	Loaded the cleaned data; validated row/field quality; analyzed customers, orders, products, countries, monthly revenue and RFM.
Python + Statistics	Used Pandas, NumPy and Matplotlib for descriptive statistics, distributions, IQR high-value analysis, RFM scoring, contribution, outliers and correlation.
Power BI	Built KPI and analytical views across Executive Overview, Customer & RFM Analysis, and Product & Market Analysis.

Method principle

Each tool had a distinct role: prepare $\rightarrow$ validate $\rightarrow$ analyze $\rightarrow$ visualize $\rightarrow$ decide.

Overall Business Performance

Revenue scale, order economics and timing

REVENUE	ORDERS	CUSTOMERS	AOV
₹17.37M	36,969	5,878	₹469.98

PERFORMANCE SCORECARD

Metric	Result	Interpretation
Revenue	₹17.37M	Overall sales value in the cleaned analysis base.
Orders	36,969	Distinct invoice-level purchasing activity.
Customers	5,878	Identified customer base for customer-level analysis.
AOV	₹469.98	Baseline for basket expansion and cross-sell.
Products	4,631	Breadth of distinct stock codes.

SALES TREND INSIGHT

Monthly revenue varies materially across the observation period. Stronger months reach roughly ₹1.16M–₹1.17M, while lower months are around ₹0.45M–₹0.52M. This variation should be incorporated into demand, inventory and campaign planning.

LOW	MID	HIGH	PEAK	PEAK
₹0.45–0.52M	~₹0.70M	~₹1.00M	~₹1.16M	~₹1.17M

MANAGEMENT INTERPRETATION

- Investigate peak months by customer segment, product mix and market to understand what drives outperformance.
- Use AOV as a baseline KPI for bundles and cross-sell initiatives.
- Do not evaluate revenue without considering customer and market concentration.

Key takeaway

The business has sufficient scale for structured growth; the next gains should come from stronger customer economics and resilience.

Customer Behavior Analysis

The average customer is not the typical customer

CUSTOMERS 5,878	MEAN REVENUE ~₹2,955	MEDIAN REVENUE ~₹868	HIGH-VALUE 633
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CUSTOMER VALUE DISTRIBUTION

Statistic	Value	What it tells us
Mean	~₹2,955	Pulled upward by high spenders.
Median	~₹868	Closer to the typical customer.
75th percentile	~₹2,248	Most customers are below this level.
Maximum	₹580,987	Very high customer value exists in the base.

Distribution insight
The mean–median gap indicates a strongly right-skewed customer revenue distribution. Customer segmentation is therefore more useful than a single average.

HIGH-VALUE CONCENTRATION

The IQR method produced an upper high-value threshold of approximately ₹5,107. 633 customers exceeded that threshold and generated approximately ₹11.34M, representing about 65.3% of total customer revenue.

TOP HIGH-VALUE CUSTOMERS

Rank	Customer ID	Revenue
1	18102	₹580,987.04
2	14646	₹528,602.52
3	14156	₹313,437.62
4	14911	₹291,420.81
5	17450	₹244,784.25
6	13694	₹195,640.69
7	17511	₹172,132.87
8	16446	₹168,472.50
9	16684	₹147,142.77
10	12415	₹144,458.37

Business implication
A small high-value cohort creates outsized financial exposure. Retention strategy should therefore be value-based, not one-size-fits-all.

RFM Customer Segmentation

From transaction history to customer treatment

RFM FRAMEWORK

Dimension	Definition	Higher score
Recency	How recently the customer purchased	More recent
Frequency	How often the customer ordered	More orders
Monetary	How much revenue the customer generated	More revenue

SEGMENT PROFILE

Segment	Customers	Approx. revenue share	Recommended treatment
Champions	1,289	~68%	Protect & reward
Loyal Customers	708	~11%	Upsell / cross-sell
Potential Loyalists	878	~5%	Increase engagement
At Risk	825	~9%	Win-back
Lost	~2,000	~4%	Selective reactivation

WHAT THE SEGMENTATION REVEALS

- Champions are the core economic engine because they combine strong recency, frequency and monetary value.
- At Risk customers are strategically important because historical value exists but recent activity has weakened.
- Potential Loyalists are a conversion pool: recent customers who have not yet developed strong purchase frequency.
- Lost customers should be approached selectively with low-cost tests rather than blanket incentives.

SEGMENT PLAYBOOK

Segment	Action
Champions	Loyalty benefits, early access, proactive service
Loyal Customers	Bundles, cross-sell and basket expansion
Potential Loyalists	Onboarding and repeat-purchase nudges
At Risk	Personalized win-back journeys
Lost	Targeted reactivation tests

Retention priority

Protect the customers who already create the most value, then recover customers whose historical value is at risk.

Product Performance Analysis

Revenue, volume and unit economics must be read together

FIVE PRODUCT LENSES

Lens	Question	Decision use
Revenue	Which products drive sales value?	Protect key products
Quantity	Which products drive volume?	Demand / inventory planning
Orders	Which products recur across transactions?	Breadth of demand
Revenue per unit	Which products create more value per unit?	Mix / pricing review
Pareto	How quickly does revenue accumulate?	Concentration monitoring

OBSERVED PRODUCT INSIGHTS

- Revenue and quantity leaders are not necessarily the same; volume is not a proxy for value.
- Revenue-per-unit adds an economic lens to the product portfolio.
- Pareto analysis was used to understand concentration among leading products.
- Service/adjustment entries such as BANK CHARGES and DOTCOM POSTAGE should be interpreted separately from ordinary merchandise.
- The largest single quantity transaction was PAPER CRAFT, LITTLE BIRDIE: 80,995 units and ₹168,469.60 revenue. The arithmetic is valid, so it was retained rather than automatically deleted.

PRODUCT EXAMPLES

Product	Revenue	Quantity	Revenue / Unit
DOORMAT	₹1,071.25	167	₹6.41
3 STRIPE MICE	₹1,241.10	662	₹1.87
50'S CHRISTMAS	₹2,272.25	1,885	₹1.21

Decision rule

Use revenue + quantity + revenue per unit together before changing inventory, promotion or assortment priorities.

Market & Geographic Analysis

The dominant market is both an advantage and a risk

MARKETS	UK REVENUE	UK SHARE	CUSTOMERS
Distinct countries	~₹14.4M	~83%	5,878

REVENUE CONCENTRATION

Market group	Revenue role	Business interpretation
UK	~83% of total revenue	Core market; strongest proven demand and highest exposure.
Other markets	~17% of total revenue	Diversification and expansion opportunity.

Approximate revenue concentration: 8 of 10 visual blocks represent the UK share

MARKET STRATEGY

Priority	Action	Rationale
1	Protect UK retention and service	Largest revenue exposure.
2	Identify 2–3 secondary markets	Diversify without abandoning the core.
3	Compare market-level product mix	Localize proven demand.
4	Track market-level retention	Measure sustainable growth.

RISK INTERPRETATION

- Geographic concentration is a resilience issue, not only a reporting KPI.
- Expansion should be evidence-led using revenue, customers, frequency and product mix together.
- Secondary-market growth can complement—not replace—the UK strategy.

Key takeaway

Keep the UK as the core market while deliberately building a broader revenue base.

Business Problems & Recommendations

Evidence → impact → action

01 CUSTOMER REVENUE CONCENTRATION

Evidence	Impact	Recommended action
633 high-value customers ≈65.3% of customer revenue	Churn can create disproportionate revenue loss.	Create a high-value retention program with loyalty, proactive service and personalized offers.

02 UK MARKET DEPENDENCE

Evidence	Impact	Recommended action
UK ≈83% of total revenue	Market disruption can materially affect total revenue.	Test secondary markets with localized product and customer strategies.

03 AT RISK CUSTOMERS

Evidence	Impact	Recommended action
825 At Risk customers; ≈9% of customer revenue	Historical value may be lost if inactivity continues.	Run personalized win-back journeys with measurable response thresholds.

04 PRODUCT CONCENTRATION

Evidence	Impact	Recommended action
Pareto analysis shows revenue accumulates among leading products.	Availability or assortment issues in key products can affect sales.	Protect leading products while monitoring concentration and maintaining portfolio breadth.

DECISION HIERARCHY

PROTECT Champions	RECOVER At Risk	DIVERSIFY Markets	OPTIMIZE Products
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90-Day Business Action Plan

Move from analysis to measurable execution

0–30 DAYS | PROTECT & RECOVER

Action	Primary measure
Launch Champion retention treatment	Retention rate / revenue at risk
Prioritize At Risk win-back	Reactivation rate / recovered revenue
Review top revenue products	Availability / revenue continuity

31–60 DAYS | TEST & LEARN

Action	Primary measure
Select 2–3 secondary markets	Revenue / repeat-order growth
Test localized campaigns	Conversion / repeat purchase
Review product mix by market	Category contribution

61–90 DAYS | SCALE WHAT WORKS

Action	Primary measure
Scale winning retention treatment	Customer value / retention
Scale strongest market tests	Incremental revenue
Formalize concentration monitoring	Monthly concentration KPI

MANAGEMENT SCORECARD

Customer	Market	Product	Commercial
Retention rate	Revenue share by country	Top-product revenue share	Revenue
At Risk recovery	Repeat orders by market	Revenue per unit	Orders
High-value revenue	Customer growth by market	Quantity by product	AOV

Success principle

Measure → act → compare → learn → scale. The dashboard should become a recurring management tool.

Technical Appendix

Validation, RFM logic and reporting definitions

DATABASE & VALIDATION

Control	Result / treatment
Final rows	779,425
NULL checks	Zero NULLs across the final nine fields
Exact duplicates	No exact duplicate rows
Quantity / Price	Non-positive transactions excluded during cleaning
Revenue validation	Revenue matched Quantity × Price
Customer / Country fields	Validated for structural quality
Cancellation / returns	Excluded through the Quantity/Price cleaning rule

RFM SCORING LOGIC

RFM was calculated at customer level. NTILE(5) created five groups for each dimension. Recency was scored so more recent customers received higher scores; Frequency and Monetary were scored so larger values received higher scores. The combined score used $R \times 100 + F \times 10 + M$.

POWER BI MODEL

The transaction table was related to the RFM customer table through Customer ID using an active many-to-one, single-direction relationship. This supported segment analysis without allowing static RFM data to create misleading cross-filtering.

CORE KPI DEFINITIONS

KPI	Definition
Total Revenue	SUM(Revenue)
Total Orders	DISTINCTCOUNT(Invoice)
Total Customers	DISTINCTCOUNT(Customer ID)
Total Products	DISTINCTCOUNT(StockCode)
Average Order Value	Total Revenue ÷ Total Orders
Revenue per Unit	Total Revenue ÷ Total Quantity

TOOLS

Excel + Power Query → preparation. MySQL + SQL → validation and structured analysis. Python with Pandas, NumPy and Matplotlib → deeper analysis and statistics. Power BI → interactive reporting and communication.

Conclusion & Next Steps

The final business story

CONCLUSION

The analysis shows a meaningful e-commerce revenue base with significant concentration across customers and geography. The strongest opportunity is to protect high-value relationships, recover valuable inactive customers and broaden the revenue base beyond the dominant market.

FIVE FINDINGS TO REMEMBER

Finding	Why it matters
₹17.37M revenue	Establishes the overall commercial baseline.
633 high-value customers ≈65.3% of customer revenue	High-value retention is financially critical.
Champions ≈68% of customer revenue	Champions deserve the strongest retention investment.
UK ≈83% of total revenue	Diversification can improve resilience.
Mean ≈₹2,955 vs median ≈₹868	Customer value is highly uneven; segmentation is necessary.

FUTURE ANALYTICAL OPPORTUNITIES

- Customer Lifetime Value (CLV) to estimate long-term customer economics.
- Retention/churn prediction to identify customers likely to become inactive.
- Market expansion scoring to prioritize countries for investment.
- Product recommendation and basket analysis for cross-sell opportunities.
- Automated refresh and recurring KPI monitoring.

Final statement

This project demonstrates an end-to-end Data Analyst workflow: prepare reliable data, validate it with SQL, analyze customer and business behavior with Python and statistics, communicate through Power BI, and convert evidence into actionable decisions.

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